Output Data Organization

1) What do “xy_data” and “scan_data” mean?

The "xy_data" belongs to structure or position dependent data, and "scan_data" directly relates with scanning variables. The Crosslightview window is defaulted with structure-dependent data. Clicking the white "material_num" menu at the Crosslightview window top, you will find all the structure-related physical quantities, like energy band, mobility, carrier concentration, electric field, potential etc. With Crosslightview window, go to the top "View" menu and choose "Scan Bias Data", you will be led to the "scan_data" category and you have to import data from certain .std file. With the popup window “Draw Relation”, you will find all the variables of “scan_data” category like current, voltage, light, time and power etc.

2) What are scanline number and dataset number and their relationship?

There are scanline number and data set number. The "equilibrium" command line is defaulted as the 1st scanline, and it would generate just one data set (1,1) by default because the simulation usually starts with "equilibrium" command line. Here the data set is in the format ***_data=(n1,n2), where n1 is the stating data set number and n2 is the ending data set number. If a particular scan line just generates one data set, then n1=n2. The "equilibrium" command will initialize all the scanning variables to zero. Each subsequent "scan" command line will be numbered internally by the software as the 2nd, the 3rd (etc.) scanline. Each "scan" command line can generate one data set or several data set. Please click any portion of a "scan" command line and read the "Wizard" explanation.

Please note that the initial variable value, which is the final "value_to" of the immediately preceeding scanline (zero if never changed since "equilibrium"), the final variable value as "value_to" in the current scanline, and the "print_step" in the current scanline all determine how many data sets will be generated. Please check with the following example with .sol file.

```
$ scanline=1 by default
equilibrium
$ above scanline generate 1 data set (1,1) by default. At this
$ moment, all scanning varibales are set to zero.

newton_par damping_step=3. print_flag=3  &&
  change_variable=no var_tol=1.e-1 res_tol=1.e-4

$ scanline=2
scan var=voltage_3 value_to=1. print_step=0.5 &&
  init_step=0.01 min_step=1.e-5 max_step=0.1
$ above scanline generate 2 data (2,3) sets because "print_step=0.5"

$ scanline=3
scan var=voltage_2 value_to=-0.5 print_step=5. &&
  init_step=0.01 min_step=1.e-5 max_step=0.2
$ above scanline generate 1 data set (4,4)

$ scanline=4
scan var=voltage_2 value_to=1. print_step=0.1 &&
  init_step=0.01 min_step=1.e-5 max_step=0.06
$ above scanline generate 15 data sets (5,19) (at "voltage_2"=-0.4, -0.3, -0.2,
$ ......,1.0) because "print_step=0.1"
```

So after the above execution, total 19 data sets will be generated, which are associated with different scanlines. One could check the data sets generated in the project directory. The file ***.sol.msg contains details of data set and the corresponding bias.

One could put large “print_step” (larger than the absolute value, |value_to=*** minus the initial value of the scanning variable|) so that each scanline just generate one data set. This will make the two numbers, i.e., scanline number and the dataset number, are coincident and one just need to note the scanline number in the .sol file.

3) How to plot scanning curves like I-V curve in .plt file by citing scanline and dataset generated in .sol file?

Although some other scanning variables are not relevant with contact pad number, such as “light”, “time” and “wavelength” etc, I-V curves are associated with contact number (for example “var=voltage_2”, and the data set number is controlled by the "get_data" command as illustrated by the following example. One could also specify the specific “scanline=num” in the “plot_scan” command line as shown below.

```
get_data  main_input=apd01ph.sol &&
  sol_inf=apd01ph.out &&
  xy_data=[4  6]  scan_data=[4  6]

plot_scan scan_var=voltage_1 variable=current_2 &&
  scale_2=log scanline=4 merge_next=yes
$ For the above command line, the scanned bias is applied onto contact pad
$ num=1, & the total current read through contact pad num=2.
$ This plot is particularly associated with scanline=4.
$ The next "plot_scan" curve will be merged together with this "plot_scan" cruve.

plot_scan scan_var=voltage_1 variable=current_2 &&
  scale_2=log scanline=6 merge_next=no

plot_scan scan_var=voltage_1 variable=current_2 &&
  scale_2=linear scanline=4 merge_next=yes &&
  data_file=IV_dark.dat
$ Above command line, vertical scale becomes linear
$ Export data into file name "IV_dark.dat"

plot_scan scan_var=voltage_1 variable=current_2 &&
  scale_2=linear scanline=6 merge_next=no &&
  data_file=IV_photo.dat
```

In the plt file, one has to use the same variable name appeared in the corresponding “scan” line in .sol file. When plotting a curve by plt file, please make sure that your variable been changed (varied) by the corresponding “scan” line in .sol file.

4) Output files are saved in file numbers like my.out_0005. How do I know which bias condition it was saved?

Please look up in my.sol.msg file.

About setting a-si trap models

5) What are the model used for setting a-si:H tail states and deep level dangling bond states?

A) Exponential decay model for tail states near conduction & valence band edges

$$DOS(E) = \frac{N_{trap}}{E_{tail}} \exp[-\frac{(E - E_0)}{E_{tail}}] = DOS_{\max} \exp[-\frac{(E - E_0)}{E_{tail}}]$$

where E_0 is the characteristics energy level specifying the conduction band edge or valence band edge depending whether the tail is at the conduction band edge or valence band edge. E is assumed to be larger than E_0 and $\boldsymbol{DOS_{max}=N_{trap}/E_{tail}}$.

Input in macro file or in sol file as below:

```
trap_conc_i (i trap number, integer) value=Ntrap in m-3
traplevel_tail_i (i trap number, integer) value=Etail in eV
```

besides setting for capture cross section for electron and hole.

B) Gaussian distribution model for dangling bond states within the band gap

$$DOS(E) = \frac{N_{trap}}{\sqrt{2\pi\sigma}} \exp[-\frac{(E - E_0)^2}{2\sigma^2}] = DOS_{\max} \exp[-\frac{(E - E_0)^2}{2\sigma^2}]$$

where E_0 is the characteristics energy level specifying Gaussian distribution center. The reference energy for E_0 is from the conduction band edge, i.e., E_0 measured from the conduction band edge.

$$DOS_{\max} = \frac{N_{trap}}{\sqrt{2\pi\sigma}}$$

Input in macro file or in sol file as below:

```
trap_conc_i (i trap labeling number) value=Ntrap in m-3
trap_level_i (i trap labeling number) value=E0 in eV
traplevel_stddev_i (i trap labeling number) value=σ in eV
```

besides setting for capture cross section for electron and hole.

6) How to use such models for a-si:H trap setting?

If user has already experimental data (or data from other sources), the end user can choose to fit the data with above models to extract relevant input parameters needed for setting in macro file or in sol file.

Alternatively, user can take the simple estimation method (less accurate) without too much fitting work.

a) For exponential decay, since $\ln[DOS(E)] - \ln DOS_{\max} = -\frac{(E - E_0)}{E_{tail}}$, the linear extrapolation would give slope={-1/E_{tail}} and vertical axis intercept=ln$DOS_{\max}$. From these one could have estimated E_{tail} and N_{trap} for input in the macro or sol file.

b) For Gaussian distribution, one could estimate either with the half-mean width w_{half} or one-decade width w_{10th} as follows depending on the visual resolution of the distribution plot.

$$\sigma = w_{half} /[2\sqrt{\ln 4}] = 0.42466 w_{half} = w_{10th} /[2\sqrt{\ln 100}] = 0.23300 w_{10th}$$

Then estimate E_0 and DOS_{max} from the plot and extract N_{trap}. In this way, all the three parameters for Gaussian distribution to be put in macro file or sol file are obtained.

The user can also construct own distribution by using fixed value for ceratin parameter(s). For example, when fixxing DOS_{max}, the exponential decay becomes one-parameter E_{tail} dependent and Gaussian distribution becomes two-parameters E_0 and σ dependent. Certainly, the other parameters can also be fixed. When all fixed, the distribution is constructed.

GaN/Si stress issue

7) What is the definition for “lattice constant” and “lattice spacing”?

The "lattice constant" is the zero-strain lattice spacing of a material and "lattice spacing" is the actual strained LED material spacing.

8) How does the strain computed in Crosslight macro file?

In Crosslight macro, there is definition of substrate lattice constant and the assumption in band structure model is to assume all layers in LED are forced to have the same lattice spacing as the substrate lattice constant and thus tensile or compressive strain for each MQW layer can be calculated accordingly.

9) What would it be if using InGaN/GaN/Sapphire and the substrate (Sapphire) lattice spacing is relaxed to GaN lattice constant?

For InGaN/GaN/Sapphire, as an example, if the substrate (Sapphire) lattice spacing is relaxed to GaN lattice constant, substrate is assumed to be relaxed and zero stress exists in substrate. In another word,
GaN_substrate_lattice_spacing=GaN_substrate_lattice_constant.
However, since
lattice_constant_InGaN>lattice_spacing_in_substrate,
InGaN MQW will have compressive strain.

10) What would it be if using AlGaN/Silicon or GaN/Silicon and the substrate (Silicon) lattice spacing is not relaxed to AlGaN or GaN lattice constant?

In this case, the method would be to estimate the substrate lattice spacing based on the amount of stress in GPa if silicon substrate bending caused the GaN or AlGaN buffer/substrate to be highly tensile stressed (+17 percent). A simple calculation can convert GPa to strain in substrate and this will allow end user to calculate the substrate lattice spacing. Since our existing model assumes whole LED_spacing=substrate_lattice_constant, user just needs to set substrate_lattice_constant=substrate_lattice_spacing_under_tensile_strain.

The above shows how to use the Crosslight conventional strain model. In the case of stress-based model, the substrate spacing is automatically set based on the substrate bending tensile stress (in GPa). The stress-based model would be more powerful.

11) How does Crosslight APSYS handle defects and dislocations?

As for defects and dislocations, APSYS does not model the dynamics of these and will only treat them as SRH recombination centers. However, if user would like to treat mechanical bending/defects-propagation, Crosslight has a separate product called CSuprem to handle such mechanical problems. APSYS is more focused on electrical/optical calculation.

<u>**Other miscellaneous questions**</u>

12) Given metal/semiconductor interface contact resistivity in unit of Ohm*m^2, how do I include this in simulation?

The best way is to use metal macro and include the contact resistivity by adding a term to the bulk resistivity value equal to rho_contact/thickness, where rho_contact is the contact resistivity and thickness is the thickness of the metal layer. Just make sure the metal work function aligns with semiconductor band edge to avoid any Schottky contact effect.

13) How can the material concentration related resistivity be included?

If the resistivity is caused by material impurity concentration, such as shown by Fig. 21 on page 32, Physics of Semiconductor Devices, 2nd edition, S. M. Sze, John Wiley & Sons, its effect can be modeled by setting concentration dependent material macros. Important material parameters affected by the impurity concentration would be minority carrier lifetime and mobility etc. For solar cell, the minority carrier lifetime value might be sensitive to affect the short-circuit current density.

14) How to get C-V plot from AC analyses?

In Crosslight device simulator, AC is usually done as spectrum plot for each data set. Here is answer on how to plot CV curve from AC analyses.

<u>In sol file,</u>
a) Set "more_output ac_data=yes" in ***.sol (eg, pn.sol) and print and save many data sets for the corresponding scanline. Please make sure that the corresponding scanline in sol file have small "print_step" to generate quite a few datasets (see <u>Output Data Organization</u> portion in this pdf file).

<u>In plt file</u>,
b) Cite the datasets range as generated by the particular scanline. Equilibrium scanline (generating dataset No 1) does not contain AC

information and should be avoided here. For example (assuming that the major scanline=2 with voltage_1 scanned has generated data sets from No=2 to No=16),

```
get_data main_input=pn.sol sol_inf=pn.out &&
    xy_data=[2 16] scan_data=(2 16)
```

c) Set AC frequency (6. for 1MHz) with "log_freq1=*.* log_freq2=*.*" with "ac_voltage……" command line;
d) Set "versus_bias=yes" with the "ac_voltage……" command line;
e) Set "scan_var=voltage_1" with "set_xydata_for_scan……" command line;
f) Set "variable=capacitance_1" with the "plot_ac_curr……" command line.

After these setting, for example, we have in plt file as follows to plot the C-V curve.

```
get_data main_input=pn.sol sol_inf=pn.out &&
    xy_data=[2 16] scan_data=(2 16)

ac_voltage log_freq1=6. log_freq2=6. contact_num=1 &&
  freq_point=2 versus_bias=yes scanline=2
set_xydata_for_scan scan_var=voltage_1
plot_ac_curr variable=capacitance_1
```